

SUMMARY OF TECHNICAL REPORT AND RECOMMENDATIONS

Since 1965 the World Health Organization (WHO) has published guidelines for the diagnosis and classification of diabetes. These were last reviewed in 1998 and were published as the guidelines for the Definition, Diagnosis and Classification of Diabetes Mellitus and its Complications. Since then more information relevant to the diagnosis of diabetes has become available. In November 2005 a joint WHO and International Diabetes Federation (IDF) Technical Advisory Group met in Geneva to review and update the current WHO guidelines.

After consideration of available data and recent recommendations made by other organisations, the Group made the following recommendations:

Recommendation 1: The current WHO diagnostic criteria for diabetes should be maintained ? fasting plasma glucose $\geq 7.0\text{mmol/l}$ (126mg/dl) or 2h plasma glucose $\geq 11.1\text{mmol/l}$ (200mg/dl). Despite the limitations with the data from which the diagnostic criteria for diabetes are derived, the current criteria distinguish a group with significantly increased premature mortality and increased risk of microvascular and cardiovascular complications.

Recommendation 2: Since there are insufficient data to accurately define normal glucose levels, the term 'normoglycaemia' should be used for glucose levels associated with low risk of developing diabetes or cardiovascular disease, that is levels below those used to define intermediate hyperglycaemia.

Recommendation 3: The current WHO definition for Impaired Glucose Tolerance (IGT) should be maintained for the present. Consideration should be given to replacing this category of intermediate hyperglycaemia by an overall risk assessment for diabetes, cardiovascular disease, or both, which includes a measure of glucose as a continuous variable.

DEFINITION AND DIAGNOSIS OF DIABETES MELLITUS AND INTERMEDIATE HYPERGLYCEMIA

Recommendation 4: The fasting plasma glucose cut-point for Impaired Fasting Glucose (IFG) should remain at 6.1mmol/l. This decision was based on concerns about the significant increase in IFG prevalence which would occur with lowering the cut-point and the impact on individuals and health systems. There is a lack of evidence of any benefit in terms of reducing adverse outcomes or progression to diabetes and people identified by a lower cut-point eg 5.6mmol/l (100mg/dl) have a more favourable cardiovascular risk profile and only half the risk of developing diabetes compared with those above the current WHO cut point. Lowering the cut-point would increase the proportion of people with IGT who also have IFG but decreases the proportion of people with IFG who also have IGT.

Recommendation 5:

1. Venous plasma glucose should be the standard method for measuring and reporting glucose concentrations in blood. However in recognition of the widespread use of capillary sampling, especially in under-resourced countries, conversion values for capillary plasma glucose are provided for post-load glucose values. Fasting values for venous and capillary plasma glucose are identical.
2. Glucose should be measured immediately after collection by near-patient testing, or if a blood sample is collected, plasma should be immediately separated, or the sample should be collected into a container with glycolytic inhibitors and placed in ice-water until separated prior to analysis.

Recommendation 6: The oral glucose tolerance test (OGTT) should be retained as a diagnostic test for the following reasons:

- Fasting plasma glucose alone fails to diagnose approximately 30% of cases of previously undiagnosed diabetes.
- An OGTT is the only means of identifying people with IGT.
- An OGTT is frequently needed to confirm or exclude an abnormality of glucose tolerance in asymptomatic people.

An OGTT should be used in individuals with fasting plasma glucose 6.1-6.9mmol/l (110-125mg/dl) to determine glucose tolerance status.

SUMMARY OF TECHNICAL REPORT AND RECOMMENDATIONS

Recommendation 7: Currently HbA1c is not considered a suitable diagnostic test for diabetes or intermediate hyperglycaemia.

The following Table summarises the 2006 WHO recommendations for the diagnostic criteria for diabetes and intermediate hyperglycaemia:

DIABETES:

Fasting plasma glucose ≥ 7.0 mmol/l (126mg/dl) OR 2-h plasma glucose ≥ 11.1 mmol/l (200mg/dl).

IMPAIRED GLUCOSE TOLERANCE (IGT):

Fasting plasma glucose < 7.0 mmol/l (126mg/dl) AND 2-h plasma glucose ≥ 7.8 and < 11.1 mmol/l (140mg/dl to 200mg/dl).

IMPAIRED FASTING GLUCOSE (IFG):

Fasting plasma glucose 6.1 to 6.9mmol/l (110mg/dl to 125mg/dl) AND (if measured) 2-h plasma glucose < 7.8 mmol/l (140mg/dl).

* Venous plasma glucose 2-h after ingestion of 75g oral glucose load.

* If 2-h plasma glucose is not measured, status is uncertain as diabetes or IGT cannot be excluded.

INTRODUCTION

Recent estimates indicate there were 171 million people in the world with diabetes in the year 2000 and this is projected to increase to 366 million by 2030. Diabetes is a condition primarily defined by the level of hyperglycaemia giving rise to risk of microvascular damage (retinopathy, nephropathy and neuropathy). It is associated with reduced life expectancy, significant morbidity due to specific diabetes related microvascular complications, increased risk of macrovascular complications (ischaemic heart disease, stroke and peripheral vascular disease), and diminished quality of life. The American Diabetes Association (ADA) estimated the national costs of diabetes in the USA for 2002 to be \$US132 billion, increasing to \$US192 billion in 2020.

Since 1965 the World Health Organization (WHO) has published guidelines for the diagnosis and classification of diabetes. These were last reviewed in 1998 and were published as the guidelines for the Definition, Diagnosis and Classification of Diabetes Mellitus. In addition, in 2003, the ADA reviewed its diagnostic criteria. While the criteria for the diagnosis of diabetes and Impaired Glucose Tolerance (IGT) remained unchanged, the ADA recommended lowering the threshold for Impaired Fasting Glucose (IFG) from 6.1mmol/l (110mg/dl) to 5.6mmol/l (100mg/dl). In view of these developments WHO and the International Diabetes Federation (IDF) decided that it was timely to review its existing guidelines for the definition and diagnosis of diabetes and intermediate hyperglycaemia.

DEFINITION AND DIAGNOSIS OF DIABETES MELLITUS AND INTERMEDIATE HYPERGLYCEMIA

Guideline Development Group was held at WHO headquarters in Geneva on November 4-6, 2005. The specific aims of the meeting were to review relevant data which addressed and could inform the review of the diagnostic criteria for:

- Diabetes
- Impaired glucose tolerance (IGT)
- Impaired fasting glucose (IFG) (in particular the 2003 ADA revised criteria).

The definition of the Metabolic Syndrome was outside the brief of this review and was therefore not considered.

In addressing these aims and in formulating recommendations, the Technical Guideline Development Group took into account the following WHO guiding values for guideline development:

- A population perspective, not primarily an individual perspective
- Scientific integrity with evidence on efficacy
- Feasibility
- Cost-effectiveness and opportunity costs
- Sensitivity to local contexts
- Transparency
- A primary audience of health policy makers

BACKGROUND & CURRENT DIAGNOSTIC CRITERIA

WHO has published four previous Technical Reports relating to diabetes in 1965, 1980, 1985 and 1999. Over this period there have been significant changes in the diagnostic criteria and classification of diabetes and intermediate hyperglycaemia.

The current diagnostic criteria used for the diagnosis of diabetes and intermediate hyperglycaemia have been in place globally for almost a decade and are widely accepted. However, in 2003 the ADA modified its recommendations resulting in discrepancies between its recommendations and those of the WHO.

Although attention has focussed on the difference in fasting plasma glucose levels for defining IFG, there are a number of important differences between the ADA and WHO recommendations which may result in differences in an individual's classification of glucose tolerance:

- Fasting plasma glucose value used to define IFG
- Inclusion of 2-h plasma glucose value in defining IFG
- Requirement for fasting plasma glucose level in defining IGT
- Fasting plasma glucose as the recommended method for diagnosing asymptomatic diabetes by ADA whereas WHO recommends the oral glucose tolerance test.

These discrepancies have implications for the individual and for population prevalence estimates.

DEFINITION AND DIAGNOSIS OF DIABETES MELLITUS

Technical questions evaluated:

- Should the current diagnostic criteria for diabetes be changed?
- How should normal glucose levels be defined?
- How should impaired glucose tolerance be defined?
- How should impaired fasting glucose be defined?
- What diagnostic tests should be used to define glucose tolerance status?

The Technical Guideline Development Group accepted the 1999 WHO "classification and aetiological" framework for diabetes.

ISSUE 1: SHOULD THE CURRENT DIAGNOSTIC CRITERIA FOR DIABETES BE CHANGED?

There are important differences between (i) defining diabetes to identify an individual with diabetes and the consequent clinical and social implications of this diagnosis and (ii) defining diabetes for epidemiological purposes. In the former the diagnosis requires careful substantiation with retesting on another day unless the person is symptomatic and the plasma glucose is unequivocally elevated whereas in epidemiological studies repeat testing is rarely performed. When repeat testing is performed, approximately 75% of people with diabetes detected in epidemiological studies are confirmed to have clinical diabetes.

In the absence of a more specific biological marker to define diabetes, plasma glucose estimation remains the basis of diagnostic criteria.

Knowledge on diagnostic cut points for diabetes has been derived from two sets of information:

- Plasma glucose levels associated with risk of diabetes specific microvascular complications, particularly retinopathy
- The population distribution of plasma glucose.

1. DIABETES COMPLICATIONS & RETINOPATHY RISK

The occurrence of diabetes-specific complications has been used to derive diagnostic cut-points for diabetes, particularly using data from epidemiological studies which have examined both prevalent and incident retinopathy across a range of plasma glucose levels.

Figure 1 shows that the prevalence of retinopathy rises markedly in the 9th decile of plasma glucose. Data from Pima Indians, a study in Egypt and unpublished NHANES III data were cited in the 1997 ADA report. Using the lower limit of the decile in which prevalent retinopathy increases significantly, the cut-points from these three studies are 10.3mmol/l, 8.6mmol/l and 8.5mmol/l respectively for 2h plasma glucose and 6.1mmol/l, 6.0mmol/l and 5.9mmol/l respectively for fasting plasma glucose.

2. POPULATION DISTRIBUTION OF PLASMA GLUCOSE

One study has reported fasting and 2-h plasma glucose cut-points and incident retinopathy. Over an 11-yr follow up, development of retinopathy increased at a baseline fasting plasma glucose cut-point of 7.0mmol/l and at a baseline 2-h post-glucose load plasma glucose of 13.3mmol/l.

A bimodal distribution of plasma glucose concentrations was first described in a 1971 study in Pima Indians and subsequently in other populations with a high prevalence of diabetes, including Pacific Islanders, Asian Indians, Mexican Americans, Egyptians and Malaysians.

ISSUE 1 RECOMMENDATION SUMMARY

In summary there are an abundance of data indicating that hyperglycaemia is harmful.

Recommendation 1: The current WHO diagnostic criteria for diabetes should be maintained ? fasting plasma glucose $\geq 7.0\text{mmol/l}$ (126mg/dl) or 2h plasma glucose $\geq 11.1\text{mmol/l}$ (200mg/dl).

Despite the limitations with the data from which the diagnostic criteria for diabetes are derived, the current criteria distinguish a group with significantly increased premature mortality and increased risk of microvascular and cardiovascular complications.

ISSUE 2: HOW SHOULD NORMAL PLASMA GLUCOSE LEVELS BE DEFINED?

One approach to addressing the issue of defining categories of intermediate hyperglycaemia is to define normal glucose tolerance.

The 2003 ADA statement defined a normal fasting plasma glucose as $<5.6\text{mmol/l}$ (down from 6.1mmol/l in 1997) and a normal 2h plasma glucose as $<7.8\text{mmol/l}$.

The 1999 WHO report used a fasting plasma glucose of $<6.1\text{mmol/l}$ and a 2h plasma glucose of $<7.8\text{mmol/l}$ as normal.

2. CLINICAL APPROACH - RISK OF ADVERSE OUTCOMES

Prospective population studies provide information on the relationship between plasma glucose and risk of death, cardiovascular disease and cancer, and of developing diabetes.

Levitan et al performed a meta-analysis of thirty eight prospective studies and confirmed that hyperglycaemia in the non-diabetic range was associated with increased risk of fatal and non-fatal cardiovascular disease.

In the Baltimore Longitudinal study, all cause mortality increased significantly from a fasting plasma glucose above 6.1mmol/l but not lower levels. For 2h plasma glucose, risk increased significantly above 7.8mmol/l.

3. PHYSIOLOGICAL APPROACH & RECOMMENDATION 2

Godsland et al reported that first phase insulin response in non-diabetic individuals begins to decrease from a fasting plasma glucose of 5.0-5.4mmol/l and late-phase insulin response declines at a fasting plasma glucose above 6.0mmol/l.

Recommendation 2: Since there are insufficient data to accurately define normal glucose levels, the term 'normoglycaemia' should be used for glucose levels associated with low risk of developing diabetes or cardiovascular disease, that is levels below those used to define intermediate hyperglycaemia.

ISSUE 3: HOW SHOULD IMPAIRED GLUCOSE TOLERANCE BE DEFINED?

In 1979 the US National Diabetes Data Group recommended the category of Impaired Glucose Tolerance (IGT) to denote a state of increased risk of progressing to diabetes, although it was also noted that many would revert to normal. This term was introduced to remove the stigma of diabetes from the other terms in use at the time to denote the range between 'normal' and diabetes. The increased risk of cardiovascular disease in people with IGT was also recognised.

VALIDITY OF THE CURRENT DEFINITION OF IGT

Data from Mauritius indicate that in people with IGT at baseline, 30% reverted to normal, 35% remain as IGT, 5% changed to IFG and 30% developed diabetes over the 11-yr follow up period.

IGT is associated with a significantly increased risk of premature mortality and cardiovascular disease.

The McMaster review reported:

- The annualized relative risk of a person with IGT progressing to diabetes was increased 6-fold compared with people with normal glucose tolerance.
- The relative risk of all-cause mortality is 1.48-fold higher in people with IGT compared with people with normal glucose tolerance.

ISSUE 3 RECOMMENDATION SUMMARY

Recommendation 3: The current WHO definition for Impaired Glucose Tolerance (IGT) should be maintained for the present.

Consideration should be given to replacing this category of intermediate hyperglycaemia by an overall risk assessment for diabetes, cardiovascular disease, or both, which includes a measure of glucose as a continuous variable.

ISSUE 4: HOW SHOULD IMPAIRED FASTING GLUCOSE BE DEFINED?

In 1997 the ADA Expert Committee introduced the category of Impaired Fasting Glucose (IFG) to describe the zone between the upper limit of normal fasting plasma glucose and the lower limit of the diabetic fasting plasma glucose (FPG 6.1–6.9mmol/l / 110–125mg/dl).

DECODE data show that of all European people with IFG defined by a fasting plasma glucose of 6.1–6.9mmol/l alone, 64.8% have isolated IFG, 28.6% have IGT and 6.6% have diabetes.

VALIDITY OF CURRENT WHO DEFINITION FOR IFG

The adoption of a cut-point for fasting plasma glucose of 6.1mmol/l (110mg/dl) by the ADA and WHO as the upper limit of 'normoglycaemia' was based on this being the level above which first phase insulin secretion is lost in response to intravenous glucose.

In 2003, the ADA revised the cut-point for IFG to 5.6mmol/l (100mg/dl).

ISSUE 4 EVALUATION OF IFG CUT-POINTS

In summary, the various approaches for deriving the most appropriate specific cut-point for defining IFG do not provide a consistent and unequivocal answer. Similar to the situation with IGT, since the rationale for IFG is to identify a risk state for future diabetes and/or future cardiovascular disease and premature mortality, a risk score combining known risk factors which includes a measure of glucose as a continuous variable, would seem a more logical approach.

SHOULD THE CURRENT WHO CRITERIA FOR DEFINING IFG BE CHANGED?

The Group considered the following points relevant to reaching its recommendation:

- Outcomes (Mortality, cardiovascular disease, microvascular complications)
- Incident diabetes
- Prevention of premature mortality and cardiovascular disease
- Impact on prevalence of IFG
- Concordance of IFG and IGT
- Economic considerations and cost implications.

PREVENTION OF PROGRESSION & IMPACT ON PREVALENCE

Unlike IGT where there is extensive evidence from well conducted randomised controlled studies that lifestyle and pharmacological interventions can prevent or delay progression to diabetes, there are currently only limited data on the preventability of progression of IFG to diabetes.

A number of studies have reported a 2-3-fold increase in IFG prevalence using the new ADA recommended criteria (5.6mmol/l) compared with WHO defined IFG (6.1mmol/l).

ECONOMIC & HEALTH POLICY IMPACT OF IFG DEFINITION

The large increase in prevalence of IFG which would result from lowering the cut-point has significant implications for the health system of many countries which are struggling to cope with caring for people with established diabetes. There is also concern that increasing the focus on IFG would divert prevention resources from IGT, where there is more extensive evidence of benefit.

RECOMMENDATION 4: IMPAIRED FASTING GLUCOSE THRESHOLD

Recommendation 4: The fasting plasma glucose cut-point for Impaired Fasting Glucose (IFG) should remain at 6.1mmol/l.

This decision was based on concerns about the significant increase in IFG prevalence which would occur with lowering the cut-point and the impact on individuals and health systems. The recommendation on IFG threshold in this report is identical to the position statement of the European Diabetes Epidemiology Group which also recommends that the diagnostic threshold for IFG should remain at 6.1mmol/l.

ISSUE 5: WHAT DIAGNOSTIC TESTS SHOULD BE USED TO DEFINE GLYCAEMIC STATUS?

1. Measurement of glucose in blood:

Measurement of glucose in blood remains the mainstay of testing for glucose tolerance status.

Glucose measured in plasma is approximately 11% higher than glucose measured in whole blood.

However this difference is dependent on haematocrit. For this and other reasons the conversion of whole blood glucose to plasma glucose is problematic and laboratory plasma measurements are recommended.

RECOMMENDATION 5 & ORAL GLUCOSE TOLERANCE TEST (OGTT)

Recommendation 5:

1. Venous plasma glucose should be the standard method for measuring and reporting. Fasting values for venous and capillary plasma glucose are identical.

2. Glucose should be measured immediately after collection by near patient testing, or if a blood sample is collected, plasma should be immediately separated, or the sample should be collected into a container with glycolytic inhibitors and placed on ice-water until separated prior to analysis.

2. Oral Glucose Tolerance Test (OGTT):

The test is recommended by the WHO. Fasting plasma glucose alone fails to diagnose approximately 30% of cases of previously undiagnosed diabetes.

GLYCAEMIC DIAGNOSTIC TESTS & PROGNOSTIC IMPLICATIONS

In summary, there is some evidence that diabetes diagnosed solely on the basis of an elevated 2-h plasma glucose is associated with a worse prognosis than diabetes diagnosed on the basis of an elevated fasting plasma glucose alone for both mortality and retinopathy. Therefore, diagnosing the 30% of individuals who have diabetes only on the basis of an elevated 2-h plasma glucose may have prognostic implications.

3. GLYCATED HAEMOGLOBIN (HBA1C) & RECOMMENDATION 6

Recommendation 6: The oral glucose tolerance test (OGTT) should be retained as a diagnostic test for the following reasons:

- Fasting plasma glucose alone fails to diagnose approximately 30% of cases of previously undiagnosed diabetes
- An OGTT is the only means of identifying people with IGT
- An OGTT is frequently needed to confirm or exclude an abnormality of glucose tolerance in asymptomatic people

An OGTT should be used in individuals with fasting plasma glucose 6.1–6.9mmol/l (110–125mg/dl) to determine glucose tolerance status.

Glycated Haemoglobin (HbA1c):

HbA1c reflects average plasma glucose over the previous 2–3 months. Taking all considerations into account, the Group concluded that the role of HbA1c in the diagnosis of diabetes and intermediate hyperglycaemia was not established in 2006 and that it could not be recommended as a diagnostic test at that time.

RECOMMENDATION 7 & TERMINOLOGY

Recommendation 7: Currently HbA1c is not considered a suitable diagnostic test for diabetes or intermediate hyperglycaemia.

Terminology:

The Group recommends using the term Intermediate Hyperglycaemia to describe glycaemic levels between 'normal' glucose tolerance and diabetes. Use of 'pre-diabetes' is discouraged to avoid any stigma associated with the word diabetes and the fact that many people do not progress to diabetes as the term implies.

APPENDIX 1: MEMBERSHIP OF GUIDELINE DEVELOPMENT COMMITTEES

WHO Guideline Steering Group: G Roglic (Chair), S Resnikoff, K Strong, N Unwin.

Technical Guideline Development Group: KGMM Alberti (Co-Chair), PH Bennett, K Borch-Johnsen, S Colagiuri (Chair), M Engelgau, P Home, Z Metelko, V Mohan, Chang Yu Pan, Qing Qiao, K Ramaiya, J Shaw, M-I Schmidt, J Tuomilehto, D Vistisen, P Zimmet.

APPENDIX 2: SUMMARY OF WHO DIAGNOSTIC CRITERIA FOR DIABETES AND INTERMEDIATE HYPERGLYCAEMIA (1965, 1980, 1985, 1999)

- Normal Fasting Glucose (1999): < 6.1 mmol/l (110 mg/dl), 2-h glucose < 7.8 mmol/l (140 mg/dl).
- Diabetes Fasting Glucose (1999): ≥ 7.0 mmol/l (126 mg/dl) OR 2-h glucose ≥ 11.1 mmol/l (200 mg/dl).
- Impaired Glucose Tolerance (IGT) (1999): Fasting glucose < 7.0 mmol/l AND 2-h glucose ≥ 7.8 and < 11.1 mmol/l (140 to 200 mg/dl).
- Impaired Fasting Glucose (IFG) (1999): Fasting glucose ≥ 6.1 and < 7.0 mmol/l (110 to 125 mg/dl) AND 2-h glucose < 7.8 mmol/l.

APPENDIX 3: COMPARISON OF 1999 WHO AND 2003 ADA DIAGNOSTIC CRITERIA

- Diabetes Fasting Glucose: WHO 1999 ≥ 7.0 mmol/l or 2-h glucose ≥ 11.1 mmol/l | ADA 2003 ≥ 7.0 mmol/l or 2-h glucose ≥ 11.1 mmol/l.
- IGT Fasting Glucose: WHO 1999 < 7.0 mmol/l (if measured) and 2-h glucose ≥ 7.8 and < 11.1 mmol/l | ADA 2003 Not required, 2-h glucose ≥ 7.8 and < 11.1 mmol/l.
- IFG Fasting Glucose: WHO 1999 6.1 to 6.9 mmol/l and 2-h glucose < 7.8 mmol/l | ADA 2003 5.6 to 6.9 mmol/l.

APPENDIX 4: DISORDERS OF GLYCAEMIA - AETIOLOGICAL TYPES AND CLINICAL STAGES

Types of Diabetes:

- Type 1 Diabetes Mellitus (Insulin requiring for survival)
- Type 2 Diabetes Mellitus (Not insulin requiring, or insulin requiring for control)
- Other Specific Types
- Gestational Diabetes Mellitus

Clinical Stages: Normoglycemia -> Impaired Glucose Tolerance / Impaired Fasting Glucose -> Diabetes Mellitus.